

Scalability & Future Plan

Date:	3 October 2026	Team ID:	UNM1611SHARVESHARAM
Project Name:	AI BlogNest API	Maximum Marks:	1 Mark

System Scaling & Evolutionary Roadmap:

This document details how AI BlogNest API can scale to accommodate larger user bases, heavy traffic spikes, and expanded multimodal AI capabilities in future iterations.

Current System Limitations:

S.No	Limitation Description	System Impact	Priority
1	Gemini Free Tier Rate Limits	15 requests per minute upper boundary can throttle high concurrent user surges.	High
2	Stateless Token Revocation	JWTs remain valid until expiration (7 days); logged-out tokens cannot be blacklisted without a cache.	Medium
3	Text-Only Generation	Current AI output generates textual content without matching cover visuals or illustrations.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Concurrency	Single Node.js instance (350 req/s)	PM2 cluster mode & horizontal pod auto-scaling (HPA) on Kubernetes.
2	Data Storage	MongoDB Atlas standard cluster	Implement MongoDB horizontal sharding across author IDs and date indexes.
3	Performance / Caching	Direct database queries for all feeds	Redis in-memory caching for trending blogs, reducing DB hits by 80%.
4	Security & Rate Limiting	Basic CORS and body size caps	Redis token-bucket rate limiter (express-rate-limit) and OAuth2 / Google SSO.

Scalability & Future Plan — Multi-Phase Product Roadmap

Future Product Roadmap (Phases 1 - 4):

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1 (Current)	Core REST Backend & Gemini 1.5 Flash User authentication, blog CRUD, AI generation, auto-categorization, and summarization.	Month 1 (Delivered)	Establishes solid backend foundation and 100% verified REST API endpoints.
Phase 2	Next.js & Tailwind CSS Frontend Dashboard Single-page web portal with live AI drafting, markdown preview, and responsive author dashboard.	Month 2 - 3	Transforms headless API into a consumer-facing SaaS publishing platform.
Phase 3	Multimodal AI Image Generation Integration with Google Imagen 3 / Stable Diffusion to automatically generate matching banner artwork.	Month 4 - 5	Delivers complete visual-plus-text articles, increasing reader engagement by 40%.
Phase 4	Social Syndication & Automated SEO Automated webhook publishing to Medium, LinkedIn, and WordPress with auto-generated meta tags.	Month 6	Enables one-click multi-platform publishing and automated search engine optimization.

Architectural Evolution Strategy:

- **Microservices Migration:** As query volume grows, decouple geminiService into an independent worker microservice with BullMQ/Redis message queues.
- **Streaming Generation:** Upgrade from HTTP JSON to Server-Sent Events (SSE) so users see AI text streaming token-by-token in real time.